Minimal friction screw frame — print A first

Status: READY for physical prototype testing; no new part has been physically tested.
Priority: P1 `A-first.3mf`; P2 optional `BC-comparisons.3mf`.
Reviewed: 25 October 2026. No deadline, scheduled printing or follow-on work.
Source: current main `1071f695b93951246a2865038e7fb9de9bdf2a05`, v3 diagnosis, E-295/E-301/E-302. This implements the owner's new request; the older diagnostic is not a prerequisite.

1. The decision

Does a shorter, low-contact frame move the same pin smoothly through 15 mm with the enclosure fully closed? Separately, do neighbors provide useful guidance without being dragged, and does horizontal printing help **functional** sliding/thread fit?

Print A-first.3mf first. It contains two frame halves and one integrated C-coupling/screw with two removable printing towers. Reuse the L motor, its successful side shim, ID3.2/OD7/0.50 mm washer, elastic bands and existing footless 40–50 mm pin with the same 4.4 mm square / lower 6 mm thread geometry. The 50 mm reference is shown in the drawing; a plain 18 mm pin is unsuitable for this frame's guide spacing. Do not print an entire replacement set.

Saved X1 Carbon project	Contents	Bambu estimate, including preparation
A-first.3mf	2 halves; 1 supported 30.6 mm rotor/screw	**2 h 20 min; 7.04 g PLA**
BC-comparisons.3mf	26 mm stand; 9 manual pins; removable boundary; close-play plate; 8 short + 4 tall height spacers; 2 upright + 2 horizontal 18 mm reference pins; 1 sliding coupon	**4 h 28 min; 10.20 g PLA**

The comparison plate is optional, independently usable and does not require nine motors. Small mass does not mean short print time: slender parts and the inherited conservative speeds/cooling dominate. Estimated times/material are slicer output, not physical measurements. The v3 *whole* plate was 4 h 59 min / 16.61 g, but included more parts; this is not an equal-content time comparison.

2. Saved printing conditions

Bambu Studio **02.08.02.61**, **Bambu Lab X1 Carbon / 0.4 mm / Generic PLA**, Textured PEI. Open the 3MF directly; no CAD export or setting entry is needed.

- Layer and first layer **0.12 mm**; nominal widths **0.45 mm**, **Arachne**, 2 walls, minimum bead 75%, minimum feature 10%. Five top/bottom shell layers; default 25% gyroid; screw, tall manual pins and orientation samples override to 100%.
- PLA **220°C**, bed **55°C**. Fan 100% after the first layer; inherited 8 s cooling threshold / 10 mm/s minimum speed. Auxiliary fan0%, exhaust70%; full settings are saved in `metadata/*-settings.json`.
- Outer/small perimeters **15 mm/s**, inner **20**, bridges **15**, first layer **15**; accelerations 1000 default, 500 outer/first, 1000 inner, mm/s². Cooling can reduce actual speed.
- Aligned seam, normal line interpolation, layer-by-layer order. XY contour, hole and elephant-foot compensation all **0**. No raft or automatic supports, including inside female threads.
- Frame halves flat back down; stand and boundary inverted on flat decks; screw upright with integral towers. Manual pins and upright coupons have 3 mm outer brims; horizontal coupons 2 mm; gap 0.12 mm. Other parts have no brim. Keep the saved poses, especially the horizontal pin's 90° roll before laying it down. Its local bridge angle is180° to cross the narrow bore; other objects retain automatic bridge angle.
- Screw tower contacts: **four 0.45 × 0.24 mm necks**, on the **external** thread. They must be clipped individually. No support contacts occur inside a pin. Contact cleanup and tall-pin adhesion remain physical risks, not slicer-certified successes.

Both archives were saved, reopened **without external profile files**, and resliced in Bambu Studio. Printer, settings, objects and overrides persisted. Both original and reopened toolpaths were checked separately; none contains automatic support or detached deposition islands. Seam/path ordering changes on reslicing, so G-code is not byte-identical. See verification and section drawing.

3. Quick test

1. Follow assembly.md. With power disconnected, fit the pin on the screw outside the frame; stop at any damaged thread or forceful fit.
2. Assemble without the motor first to observe both frame thrust stops. Then install the motor and check the same free axial float, PCB/wire clearance and full closure. The C fit must permit axial sliding; roof clearance alone is insufficient.
3. Turn gently at low, +3, +7.5 and +15 mm, both ways. Observe guides, screw bow, rotor contact and coupling stability. Repeat fully closed and briefly loosened. **A mechanism that needs loosening fails.** No deliberate lateral load or added weight during powered testing.
4. Compare with v3 using the same motor, pin, washer, shim, material and manual procedure. New screw length also changed, so improvement cannot identify one original rubbing surface by itself.
5. Only after smooth hand motion and no shaft movement/gear creep: a few slow 3 mm powered cycles, then up to the manually verified range, maximum 15 mm. Use the existing controller only when actual driver settings are known. No IRUN increase or firmware changes.
6. On the second plate, compare equal/staggered heights, missing neighbor, both play settings and boundary on/off. Then remove spacers and lay the stand on its side for **freely sliding neighbors in both directions**. This distinguishes neighbor drag from fixed-wall geometry. See assembly instructions.
7. Compare both copies of each orientation using the same bare reference screw and guide coupon. Test every exterior face. Reject damaged or catching internal threads regardless of a smooth-looking outside.

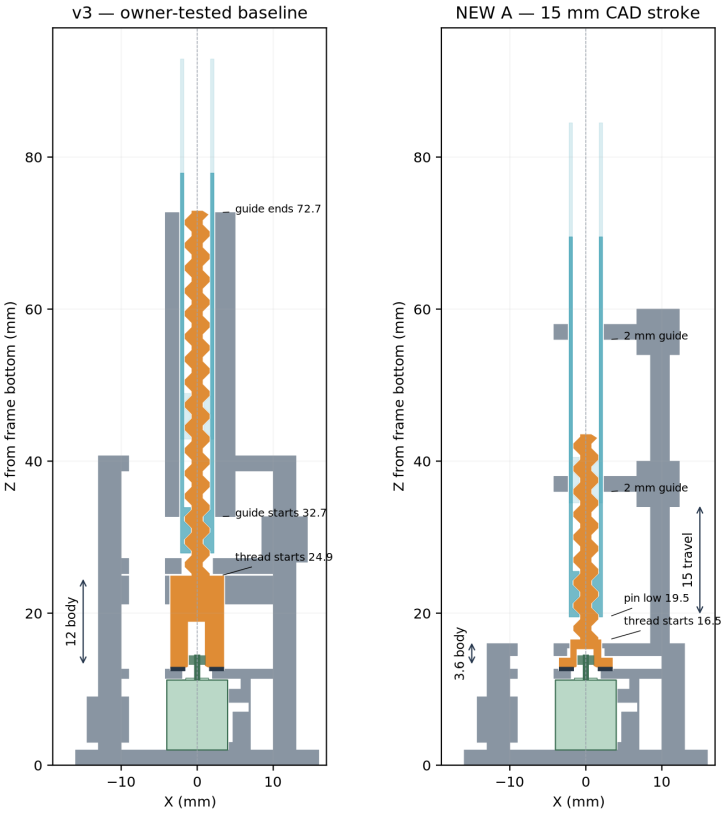
Stop: any binding, visible shaft displacement, gear migration/disengagement, rotor witness marks, pin bow, rubbing on PCB/wires, damage, support-removal scars that catch, or need to force/loosen the mechanism. Do not find a mechanical limit by running into it. Fill results.md; blanks mean unknown.

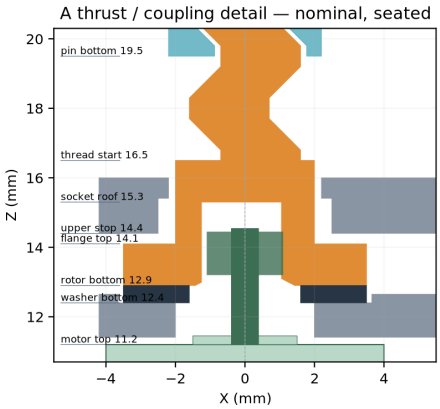
4. Interpretation

Physical result	Decision supported
A stays smooth closed at low and +15, coupling stable	Compact stack and local guides merit retention for this tested load/travel only.
Smooth without motor; binds with motor	C sliding/axis alignment remains limiting; do not compensate with current.
Smooth low; binds high	Record guide station, bow and witness marks. CAD clearance has not established usable high travel.
Rotor rubs the 0.6 mm land or shaft moves	No-contact hypothesis failed under that condition; no powered/load qualification. A deliberate short radial bearing or compliant torque interface would need a separate owner decision.
Neighbor motion drags other pins / loss beside a gap	Neighbor-only guidance is unsuitable for that state; fixed/peripheral guidance remains necessary.
Horizontal outside feels smoother but thread catches	Retain upright functional pins.
Both horizontal copies retain thread fit and slide better	Orientation is promising locally; it does not qualify a 50 mm horizontal pin or production array.

Dimensioned axial section

Minimal screw frame • section through actual meshes • 25 October 2026





Axial feature (mm)	v3	New A
Motor top / L gear	11.2 / 13.2–14.45	same
L shaft tip	14.55	14.55
C straight zone	13.15–18.9	13.15–15.3
Lower thrust / flange top	12.9 / 24.9	12.9 / 14.1
Upper stop underside	25.2	14.4
Rotor axial float	0.30	0.30
Thread start / end	24.9 / 72.9	16.5 / 43.5
Pin lower thread at low	27.9–33.9	19.5–25.5
At +15 (float excluded)	42.9–48.9	34.5–40.5
Thread overlap at 0 / 15	6 / 6 (long)	6 / 6
Guide stations	32.7–72.7	36–38; 56–58
Frame height	72.7	60.0
Radial land length / gap	3.7 / 0.20	0.6 / 0.20*

* Loose emergency land at Z15.4–16.0; no nominal radial contact.
May react misalignment. It does not guarantee zero motor shaft load.

Gray: frame Orange: rotor/screw Blue: pin (faint = +15 mm) Green: approximate L motor Dark: reused washer
CAD clearance and toolpaths are digital evidence. Physical fit, friction, shaft load and durability remain untested.

Assembly and physical comparison

A — one motor / one pin

Use **variant L**, gear approximately 2.0–3.25 mm above the metal motor top. K gives only about 0.5 mm minimum straight engagement at full float and is **not released for powered testing**. No need to remeasure the already recorded motor dimensions. The real C fit, tab/glue variation and absence of shaft motion still require observation.

1. Keep the screw and towers upright during cleanup. With flush cutters, support the screw locally and cut the **four thin necks**, alternating sides; do not twist/pull a tower. Remove the brim. Look for cracks and raised nubs on each external thread contact. Do not shave the flange, washer seat or C ribs. If a nub catches the pin, stop rather than force it through.
2. Reuse the already cleaned 40–50 mm pin: 4.4 square, short lower internal thread, open upper square bore. The wide old sacrificial foot must already be absent; it is not a functional stop in this design. A pin retaining that 12 mm foot does not fit the intended travel. The supplied optional clean pin mesh has no such foot. Do not cut off any functional guide or stop.
3. First fit the pin onto the bare screw, without the motor/frame. Check several screw positions through 15 mm of rise. The original profile/lead is retained, but tower removal and printing can still spoil it. Use fingertips only.
4. Put the washer flat in the lower pocket. Seat the rotor flange on it. Before installing the motor, close the halves and gently alternate downward/upward rotor reaction: washer/deck below and captured flange/stop above must react the load. Nominal rotor lift is **0.30 mm**. Both thrust surfaces must permit rotation. This also separates frame drag from motor/C-fit drag.
5. Open again; seat the L metal body at Z2, PCB to the open **−X side**, wires outside the ring/rotor at $X \leq -4.4$. Retain the tested shim choice; use a thinner existing shim if the halves cannot seat. Do not squeeze the PCB, glue or solder. No new spring bridge is present.
6. Lower the C socket over the gear without forcing. Confirm rotor-to-washer seating and small axial float remain possible **with** the motor. Watch that the gear/shaft does not move with the rotor. If the C fit sticks axially, the shaft can carry frictional thrust despite an open cavity roof: stop.
7. Both new halves belong together. Turn the back over by rotation, not reflection: its motor opening and PCB channel align with the front, and all three pegs meet their pockets. Use existing bands around the grooves near Z7, 36 and 56. Keep wires outside the upper bands. Close until the hard split faces meet; do not crush the frame.
8. Put the pin in the two short square guide stations. Low pin bottom is **Z19.5** with rotor seated; the thread extends Z16.5–43.5. For a 50 mm pin the tip is Z69.5, **9.5 mm above the 60 mm frame**; for a 40 mm pin it is Z59.5. Mark the initial tip position with a removable paper reference. Thread travel +3/+7.5/+15 mm gives corresponding tip rise; float can add 0.30 mm.
9. Turn the exposed screw just below the pin gently with fingertips. Check each direction at low, +3, +7.5, +15; five manual return cycles are sufficient for this comparison. Keep fingers from pushing the pin sideways. Record whether motion changes when the bands are briefly loosened; restore normal closure before powered testing.
10. Only after manual acceptance, run a few short slow cycles with your existing hardware/software and known actual settings. Start with 3 mm. Do not use time alone as proof of travel; observe the reference mark. Maximum authorized prototype range here is **15 mm**, not 40 mm. Disconnect at any obstruction; do not raise IRUN.

There is no safe crash stop at +15. Thread geometry permits further motion, but the lower guide begins losing overlap beyond about +16.2 mm at maximum float. Stay inside the marked range. The minimum 6 mm engagement retains **3 mm nominal thread-end margin** at +15. No load rating, speed or durability is established.

B — 3×3 neighbors at 5.08 mm pitch

The nine identical manual columns are deliberately **not screw driven**. Their upper 30 mm is 4.4 mm square; round Ø2.4 stems permit rotation and independent translation. They test exterior contact. They do not test screw self-locking or an entire unsupported array.

1. Turn the stand over: its flat deck is now at Z20–22 and its four legs rest on the table. Insert all nine stems. Identify the center with a marker. The center stem hole is Ø4.8, giving 1.2 mm radial clearance and no antirotation; neighboring holes Ø3.6 give nominal 0.6 mm radial play. The holes do **not** intersect.
2. Put one short spacer around each outer stem below its square body. Because spacers contact the tapered transition, the nominal low square-body bottom is **Z27.25**, not Z25. This is an assembly calculation, not a precision gauge. The center is held manually at a matching level. All nine top faces should initially be level.
3. For the **fixed-neighbor geometry condition**, lightly hold the chosen neighbors in place. Raise/lower and twist the center without forcing. This condition only tests geometry; fixed neighbors exaggerate restraint compared with a real moving array.
4. Add a tall 15 mm spacer beneath the short spacer on four alternate outer stems. Their square bodies rise 15 mm, leaving 15 mm mutual overlap. Test center travel and twist again. Remove a cardinal neighbor, then a corner neighbor, to expose boundary/gap behavior. Record which change matters.
5. Lift all pins out to add/remove the optional Ø3.0-hole liner. Its top raises the datum **1.2 mm**, but the imposed height difference remains 15. Neighbor stem radial play becomes 0.3 mm; the center opening remains Ø4.8. Check stems slide before comparing body friction. The loose fixture intentionally permits grid errors that may consume the 0.68 mm face gaps.
6. The square boundary ring stands on four legs on the deck, opening **15.0 ×15.0**, ring Z32–34. With all outer pins low, nominal outer face clearance is 0.22 mm. The ring limits array spreading. Raised neighbors can leave this ring entirely: that loss is an intended test condition, not hidden full-height support. Test ring on and off.
7. For **movable neighbors**, first leave them unheld upright on their spacers and observe whether a rising center lifts one. Then remove spacers, lay the stand on its side, position the bodies around their low positions and slide the center ± a small distance while every neighbor can move freely in either axial direction. Keep all stems through the deck and the square bodies off the table. Test with/without boundary, then stagger bodies by 15 mm. This horizontal fixture condition removes gravitational height holding; report it separately from the upright test. Touching/falling pins or stem/deck rubbing invalidate a neighbor-only friction conclusion.

No replacement pins of different width are mixed in. Both play settings retain 5.08 mm *nominal centers*; the round clearance holes intentionally allow departures from those centers. Pin-to-pin overlap disappears at a 30 mm height difference for these 30 mm bodies; a real 50 mm pin would lose it at 50 mm difference, before considering lower fixed structures.

C — same pin geometry, different print orientation

Two 18 mm upright and two identical 18 mm horizontal reference pins are provided. Horizontal means a **90° roll around the pin's long axis, then lying flat**; retain the saved pose. They are short hand-test coupons, not replacements for A's long pin.

Remove external brims only; no internal supports are generated. Identify U1/U2/H1/H2 with a marker. Inspect the open thread and the upper-bore roof. Use the same existing reference screw (or the new bare screw off the motor) for all four: start gently, establish the same engagement, and move 3 mm in each direction five times. Stop if a bridge sags or thread catches; do not ream it into a pass.

Then lay each in the U-shaped sliding coupon, same face downward, and slide along it with the same light fingertip pressure. Turn through all four faces and repeat. The open coupon isolates sliding; it is not claimed to constrain every motion. Record thread fit, straightness, seam snagging, face-dependent friction, cracking, and agreement between duplicates. Only a combined thread/sliding improvement supports a horizontal-print preference. A nice exterior alone does not.

Results — blank means not tested

Date: ___ Project filename/hash: ___ Printer/nozzle: ___ PLA: ___ Actual printed settings and changes from saved project: ___ Print succeeded / partially / failed / not printed: ___ Actual time/material if known: ___ Cleanup, adhesion or tower damage: ___

A — compare with v3

Motor L / other: ___ Shim: ___ Washer: ___ Pin length: ___ Existing pin's foot absent: ___ Screw/pin hand fit outside frame: ___ Motor secure with frame closed: ___ PCB/glue/wires free: ___ Rotor seats on washer: ___ Upward stop effective: ___ Axial float present without / with motor: ___ Gear/shaft movement: ___

Assembly / state	Upward	Downward	Wobble / bow	Rotor / frame marks	Guide contact	Closed vs loosened
v3 low						
v3 +3 / actual ___						
v3 high / actual ___						
A low						
A +3						
A +7.5						
A +15 or verified limit ___						

Coupling slips / creeps / disengages: ___ Number of hand cycles: ___ Powered test performed only after manual pass: yes / no Actual verified Vref / IRUN / controller evidence: ___ Commanded speed vs observed movement: ___ Powered cycles and measured travel: ___ Wear, cracks, unexpected contacts or stop reason: ___ Preferred assembly and why: ___

B — mutual guidance

Stem fit before body contact: ___ Liner off/on: ___ Ring off/on: ___

Condition	Center moves / twists / tilts	Neighbors disturbed	Resistance / contact location
Equal height, neighbors held (geometry only)			
Alternate +15 mm, neighbors held			
Missing cardinal / corner neighbor			
Upright, unheld neighbors resting on spacers			
Sideways, freely sliding, equal height			
Sideways, freely sliding, staggered height			
Closer-play liner / no liner			
Boundary ring / no boundary ring			

Lost overlap / collision / movement of other pins: ___ Mutual guidance useful under which conditions? ___ Failed conditions: ___

C — orientation

Same reference screw and material for all: ___

Sample	Print / thread quality	Supports or roof scarring	5 hand cycles / both directions	Sliding each face / seam	Straightness / strength
U1					
U2					
H1 (90° roll)					
H2 (90° roll)					

Repeatable preference, considering thread AND outside: ___ Optional photos or notes: ___

Interpretation after owner feedback: unknown. No physical fit, friction reduction, shaft-load rating, holding force, lifetime, full-array performance or 40 mm travel has been measured by this delivery. No next work is automatically started.